

Assignment 1

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Abstract

In recent years the role of the gut microbiome in health has been elucidated Ross et al., 2024.

1 Introduction

In this assignment we try to compare different *regressors* for *BMI* depending on the prevalence of and diversity of species of the gut micro-biome.

1.1 Body mass index

The body mass index (BMI) is a metric derived from the mass, weight and height of a person in $\frac{kg}{m^2}$. The world health organization (WHO) “World Health Organization (WHO)”, n.d. has an official categorization of different values of the BMI per age group. The BMI can be a useful estimator for ones physical condition and health Armet et al., 2022.

1.2 Gut microbiome

The gut microbiome is the sum of populations of microorganisms that live in the digestive track. This term may refer to only the populations on a binary, exists/does not exist scale but here we refer to it as not only existence but also quantification of populations of microorganisms. The importance of the presence of these colonies has to do with their role in adding elements or altering elements of the sum of all metabolic pathways originating from the digestive track.

2 Methods

2.1 Implementation

All results were created in the *Python* programming language “3.13.2 Documentation”, n.d.

2.2 Models

Three different *regressors* were used in the analysis.

Models:

1. Bayesian Ridge
2. Support Vector Regression
3. Elastic Net

Models were imported from the *sklearn* “scikit-learn: machine learning in Python — scikit-learn 1.6.1 documentation”, n.d. library in *Python* “3.13.2 Documentation”, n.d.

2.3 Evaluation metrics

Four evaluation metrics were chosen

1. Root mean squared error
2. Mean absolute error
3. R2 score
4. Median absolute error

The Root mean squared error, Mean absolute error, R2 score were chosen as evaluation metrics because they are listed as mandatory and the Median absolute error because we have seen the distributions of the pairs of features and we know:



1. The categorical data is not going to be in the final data, but in the final metadata.
2. Some sets of features contain what seem like outliers.
3. We have no median based metric till now to compensate

As a result of knowing about outliers a median metric must be included to compare with the non-median metrics.

2.4 Confidences

1000 iteration *bootstrapping* was done on models for evaluation along with repeated *k-fold cross validation* for 50 folds, 20 iterations.

The *sklearn* library “scikit-learn: machine learning in Python — scikit-learn 1.6.1 documentation”, [n.d.](#) does not contain a *bootstrapping* function that we can use to call on our models. A protocol was written at [our source files](#).

Three distinct *bootstrap* implementations were assessed.

1. Only altering the training set
2. Only altering the validation set
3. Altering both the training and validation set

For all models all three were used.

All files can be found at [this repository](#).

2.5 Figures

Figures were generated from the *matplotlib* “Matplotlib documentation — Matplotlib 3.10.1 documentation”, [n.d.](#) library, all implementations are available at [our source files](#).

In all box plots the dotted green line denotes the mean value and the orange line denotes the median.

2.6 Data transformations

Categorical values like the *Sex* category were cast to categorical numerical values.

Since the data was not z-scale normalized a z-scale normalization was done on the micro-organism colonies.

2.7 Feature selection

Columns such as *Host age* were removed from the final data before baselines were established

List of removed features:

1. Unamed 0
2. Experiment type
3. Disease Mesh ID

Then the features:

1. Project ID
2. Sex
3. Host age

Were considered metadata for our models and a baseline was created with these



in mind as was without them.

Once the baseline was established the feature selected models used a principal component analysis to keep 95% of the total variance. This left only 95 features in the dataset.

2.8 Hyper parameter optimization

The library *optuna* “Optuna - A hyper-parameter optimization framework”, [n.d.](#) was used to search for model best parameters.

3 Results

We will only focus on the baseline model that was chosen after all the different baselines were evaluated, since multiple baseline models using the metadata as it was deemed (2.7) were constructed. For more reading on the baseline models that were not chosen and used here reference the [source files](#).

3.1 Baseline

Baseline predictions:

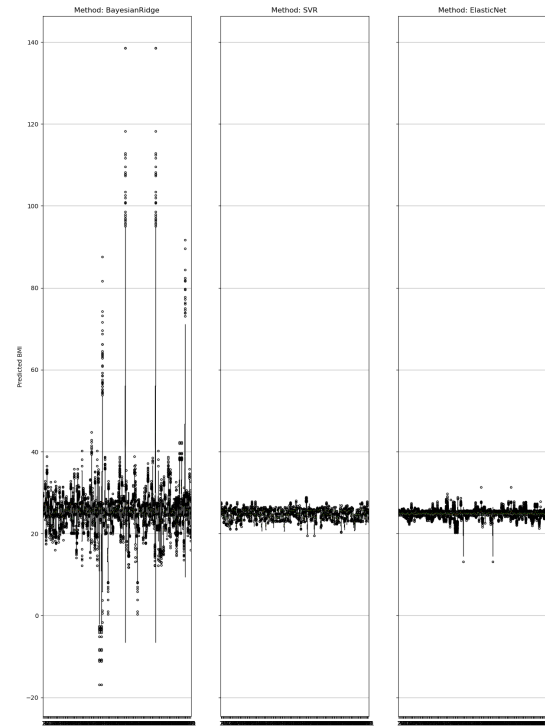


Figure 1: Bootstrapping predictions for both the development and the validation set of the baseline model

Our baseline evaluation metrics:

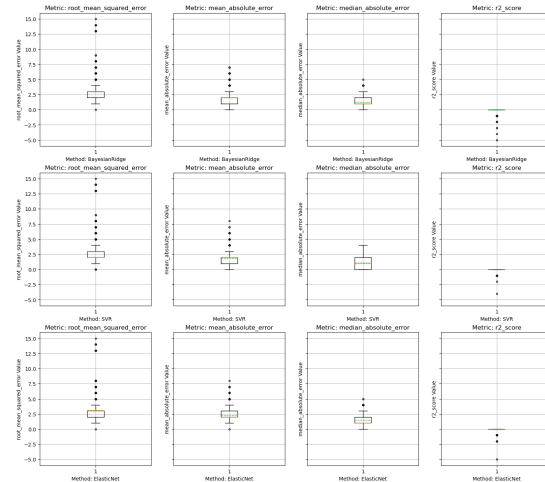


Figure 2: K-fold cross validation for the chosen baseline model

The models of Figure 1, Figure 2 is in the [models](#) folder under the name [model]_Baseline_just_species.pkl.



One thing to notice about this baseline Bayesian Ridge model is the upper and lower bound of the predictions it produces evident in Figure 1. The highest predicted bootstrap iteration is up to ≥ 150 BMI and the lowest is ≤ -10 . Not only are negative values of the BMI not allowed but values that range ≤ 10 are very unlikely to impossible. There is also something evidently wrong with the model according to the k-fold cross validation results Figure 2 as the r^2 score is negative which is indicative of poor guessing.

The other two *regressors* suffer from a separate problem, one that might render them uniquely worse than the Bayesian ridge, that being that the prediction range is very narrow and misses a large portion of the validation set and even the development set. Their metrics may not be too altered as a result of this since the predictions within range are plentiful and accurate. This is something to keep note of.

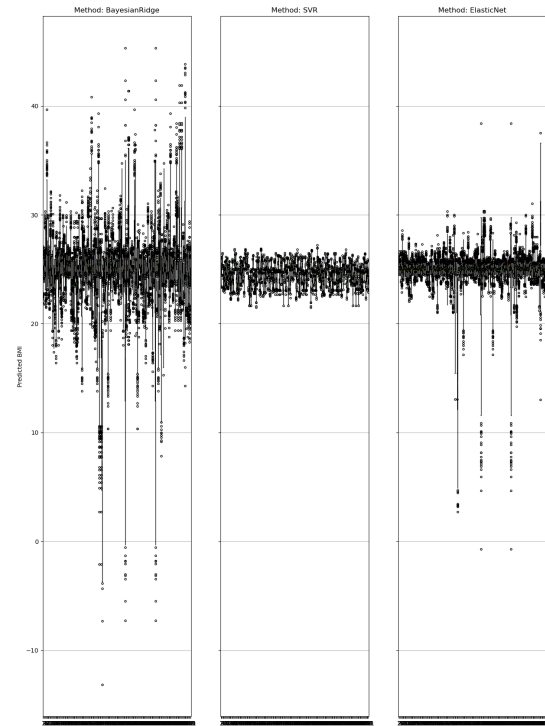


Figure 3: Bootstrapping predictions for both the development and the validation set of the feature selected model

3.2 Feature selection

A function to compare the evaluation metrics of two consecutive models has been written in [our source files](#) for a more analytical comparison of the steps we take. The function takes two distributions of evaluation metrics and calculates the difference of the two distributions, resulting in one final distribution. This final output will be a 0 mean unit variance Gaussian should there be no difference between the two models.

Feature selected model predictions:

Evaluation metrics:

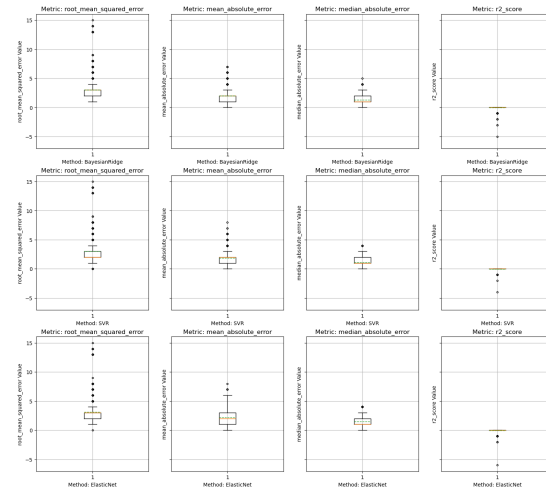


Figure 4: K-fold cross validation for the feature selected model

We will not show the difference of Gaussians here for brevity's sake, one can browse extra materials at [our notebooks](#).



These results have better consistency when it comes to the bounds of the predictions (the maximum and minimum prediction of all models but mainly the Bayesian ridge). The evaluation metrics imply that this is not just a visual difference but that the models have performed better on the data once their inputs have been processed with a feature selection method 2.7. The difference is not substantial but it is notable. One thing to notice is that the upper predictions is the one that has changed the most from the previous models. There is a tendency thus far to predict lower values rather than higher ones. This time we miss the highest BMI (being 60) in the validation set in the prediction range. This is something that cannot be helped since the data point in question is of low representation and only in the validation set.

Now that the bounds of all models are shorter we can see that the Elastic net seems to be on average not capturing the distribution properly but is sometimes (some bootstrap iterations) capturing the actual range of the dev set on the lower bound.

3.3 Hyper parameter optimization

Feature selected hyper parameter optimized predictions:

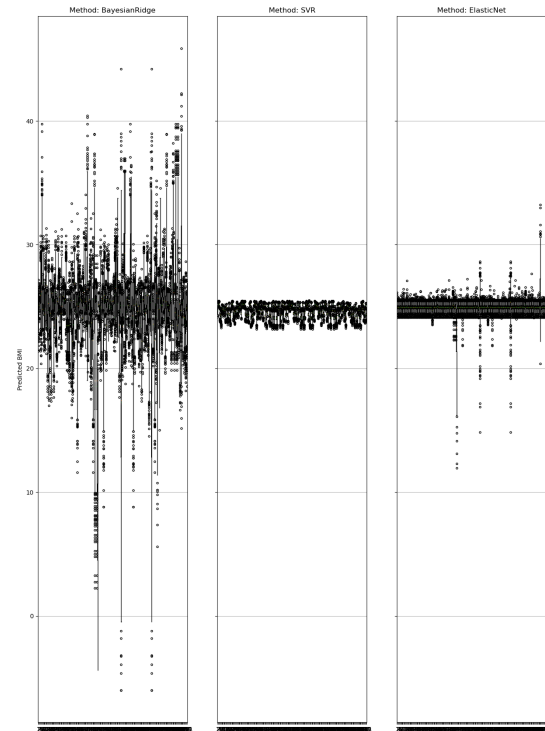


Figure 5: Bootstrapping predictions for both the development and the validation set of the feature selected optimized model

Feature selected hyper parameter optimized evaluations:

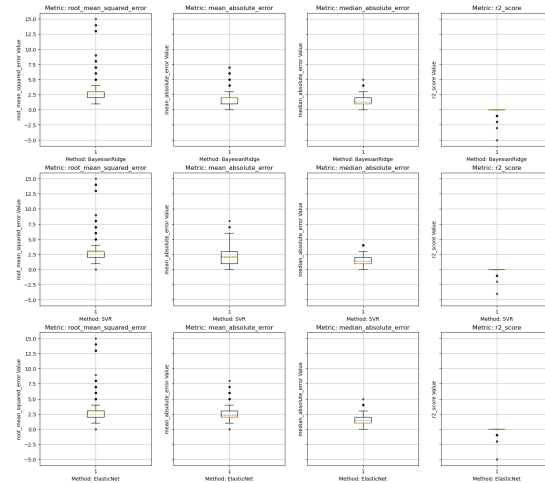


Figure 6: K-fold cross validation for the chosen feature selected optimized model

The Bayesian ridge does not seem to have been affected by the optimization parameter change. The other two models



seem to have even more narrow distribution around the mean of their predictions after the optimization. The metrics have improved but when choosing a best model the distribution range of all prediction outputs cannot be ignored and for that reason the Bayesian ridge has been chosen as the "Winner" model from the analysis. The improvements of the metrics as minor, as they were in the feature selection 3.2.

4 Conclusions

Our final "winner" model seems to have a good distribution of its predictions for the training set that was used, its metrics leave something to be desired however considering that each step did not improve them much. Seeing as the r^2 score is negative we cannot say that this model does "good" predictions. It would seem that this problem is not solvable to a satisfying degree with the given data that has been offered as of this assignment.

5 Disclaimer

Large language models (LLM's) were used during the assignment.

5.1 LLM

Open AI: Chat GPT 4.0 "ChatGPT", [n.d.](#)

5.2 Purpose

1. To check ideas, like if the z-scaled data must be zero mean and unit variance (apparently it is) or if it's conditional (or to explain ideas).
2. To ask if concepts already exist in some dependency, as was the case for the way we cast the labels of the sex

feature in the original data to binary representations (also re-used for the Project ID).

3. For acquisition of the relative documentation.
4. For helping with the L^AT_EXtable formatting in this report for time efficiency.

6 Citations

References

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